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FOREARM GRIP FOR A FIREARM

Abstract

A firearm attachment for supporting a firearm and mounting accessories to the firearm includes a handguard which is shaped to mount to a barrel of the firearm and a monopod which is removably coupled to the handguard. The monopod is removable from the handguard and may have a tool attached to it so that the monopod may be used as a handle for the tool. A flashlight and a laser emitter may be removably attached to the handguard to illuminate an area in front of the firearm and facilitate aiming the firearm. The flashlight and the laser emitter may also be removable to be usable independently of the handguard and the firearm.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] Not Applicable

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] Not Applicable

THE NAMES OF THE PARTIES TO A JOINT RESEARCH AGREEMENT

[0003] Not Applicable

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ON A COMPACT DISC OR AS A TEXT FILE VIA THE OFFICE ELECTRONIC FILING SYSTEM

[0004] Not Applicable

STATEMENT REGARDING PRIOR DISCLOSURES BY THE INVENTOR OR JOINT INVENTOR

[0005] Not Applicable

BACKGROUND OF THE INVENTION

(1) Field of the Invention

[0006] The disclosure relates to firearm attachments and more particularly pertains to a new firearm attachment for supporting a firearm and mounting accessories to the firearm.

(2) Description of Related Art Including Information Disclosed Under 37 CFR 1.97 and 1.98

[0007] Myriad attachments are known to the prior art which may be attached to the barrel and used to facilitate holding or supporting the firearm, illuminate an area in front of the firearm, or aid in aiming the firearm. But the prior art fails to describe such an apparatus which includes a pivotable, removable monopod and a tool deployable from the monopod such that the monopod may be used as a handle for the tool in separation from the handguard. The prior art also fails to describe such an apparatus which includes a removable flashlight and a removable laser light emitter.

BRIEF SUMMARY OF THE INVENTION

[0008] An embodiment of the disclosure meets the needs presented above by generally comprising a handguard configured to mount to a barrel of the firearm and a monopod removably coupled to a bottom side of the handguard via a monopod fastener. The monopod is elongated between proximal and distal ends relative to the handguard. The monopod is movable between an extended position and a retracted position. When the monopod is in the extended position, the monopod extends downwardly away from the handguard.

[0009] There has thus been outlined, rather broadly, the more important features of the disclosure in order that the detailed description thereof that follows may be better understood, and in order that the present contribution to the art may be better appreciated. There are additional features of the disclosure that will be described hereinafter and which will form the subject matter of the claims appended hereto.

[0010] The objects of the disclosure, along with the various features of novelty which characterize the disclosure, are pointed out with particularity in the claims annexed to and forming a part of this disclosure.

Description

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWING(S)

[0011] The disclosure will be better understood and objects other than those set forth above will become apparent when consideration is given to the following detailed description thereof. Such description makes reference to the annexed drawings wherein:

[0012] FIG. 1 is a top perspective view of a firearm attachment according to an embodiment of the disclosure.

[0013] FIG. 2 is a bottom perspective view of an embodiment of the disclosure.

[0014] FIG. 3 is an exploded view of an embodiment of the disclosure.

[0015] FIG. 4 is a side view of an embodiment of the disclosure which shows a monopod in an extended position.

[0016] FIG. 5 is a side view of an embodiment of the disclosure which shows a monopod in a retracted position.

[0017] FIG. 6 is an exploded view of an embodiment of the disclosure and a pair of monopods with attached tools.

[0018] FIG. 7 is an in-use view of an embodiment of the disclosure.

DETAILED DESCRIPTION OF THE INVENTION

[0019] With reference now to the drawings, and in particular to FIGS. 1 through 7 thereof, a new firearm attachment embodying the principles and concepts of an embodiment of the disclosure and generally designated by the reference numeral **10** will be described.

[0020] As best illustrated in FIGS. 1 through 7, the firearm attachment **10** generally comprises a handguard **12** which has a tubular shape for receiving a barrel **112** of a firearm **110** therein. A front end **14** and a back end **16** of the handguard **12** are open into an interior space of the handguard **12**. The barrel **112** of the firearm **110** inserts into the interior space of the handguard **12** via the back end **16** and extends out of the front end **14** when the handguard **12** is attached to the firearm **110**. Upon receiving the barrel **112** into the interior space of the handguard **12**, the handguard **12** may be attached to the firearm **110** via threaded fasteners, a snap connection, a latch, or other suitable fastening means. A rail **22** is attached to a top side **18** of the handguard **12** which may be used to mount accessories to the firearm **110**. The rail **22** extends between the front end **14** and the back end **16** of the handguard **12** and defines a plurality of grooves **24** which extend transversely with respect to a front-to-back axis of the handguard **12**.

[0021] A monopod **26** for supporting the barrel **112** of the firearm **110** on a support surface is coupled to a bottom side **20** of the handguard **12**. The monopod **26** is elongated between proximal and distal ends **28**, **30** relative to the handguard **12**. The monopod **26** is pivotable at a point near the proximal end **28** between an extended position **38** and a retracted position **36**. In the extended position **38**, the monopod **26** extends away from the bottom side **20** of the handguard **12**. In the retracted position **36**, the monopod **26** extends forwardly along the handguard **12**. The distal end **30** tapers to a point. The monopod **26** is also sized to be grasped by a user's hand when in the extended position **38**. A front surface **32** of the monopod **26** faces forwardly relative to the handguard **12** when the monopod **26** is in the extended position **38**. The front surface **32** defines a plurality of recesses **34** sized and configured to receive the user's fingers therein, thereby enhancing the user's grip on the monopod **26**.

[0022] A monopod fastener **40** removably couples the monopod **26** to the handguard **12**. The monopod fastener **40** comprises an insertion member **42** coupled to the handguard **12** and a receiving member **44** coupled to the monopod **26**, wherein the insertion member **42** is receivable into the receiving member **44**. The monopod fastener **40** further comprises a latch member **46** which is coupled to the monopod **26** and is biased via a spring **48** or other suitable member to interlock with the insertion member **42**. The latch member **46** protrudes through the front surface **32** of the monopod **26** such that the latch member **46** may be actuated against bias to release the insertion member **42** and allow removal of the monopod **26** from the handguard **12**. In other embodiments, the monopod **26** may be attached to the handguard **12** via a threaded fastener, a snap connection, a magnet, or another suitable fastening member.

[0023] A flashlight module **50** and a laser emitter module **62** are removably mounted to the handguard **12** by a flashlight mount **74** and a laser emitter mount **80** respectively. The flashlight mount **74** and the laser emitter mount **80** are positioned opposite each other across the monopod **26**. The flashlight module **50** and the laser emitter module **62** attach to the handguard **12** so they emit light forwardly with respect to the handguard **12**. The flashlight mount **74** comprises a flashlight receiver **76** which receives the flashlight module **50** therein and a flashlight magnet **78** mounted to the flashlight module **50** that secures the flashlight magnetically in the flashlight receiver **76**. The

flashlight mount **74** may have another magnet affixed to the flashlight receiver **76** or comprise a ferromagnetic material such that the flashlight magnet **78** is magnetically attracted to the flashlight receiver **76**.

[0024] Similarly, the laser emitter mount **80** comprises a laser emitter receiver **82** which receives the laser emitter module **62** therein and a laser emitter magnet **84** mounted to the laser emitter module **62** that secures the laser emitter magnetically in the laser emitter receiver **82**. The laser emitter mount **80** may have another magnet affixed to the laser emitter receiver **82** or comprise a ferromagnetic material such that the laser emitter magnet **84** is magnetically attracted to the laser emitter receiver **82**.

[0025] The flashlight module **50** comprises a flashlight housing **52**, an illumination light source **54**, a flashlight power supply **56**, a flashlight switch **58**, and a flashlight charging port **60**. The flashlight power supply **56** comprises a flashlight battery but may comprise a capacitor or other suitable power source. The flashlight switch **58** is electrically coupled to the flashlight power source and the illumination light source **54** and is operable to activate the illumination light source **54** to emit an illuminating light forwardly of the handguard **12** when the flashlight module **50** is mounted to the handguard **12**. The flashlight module **50** may also be used as a typical flashlight separate from the handguard **12**. The flashlight charging port **60** is mounted to the flashlight housing **52** and is electrically coupled to the flashlight power supply **56** to facilitate charging the flashlight power supply **56**. The flashlight charging port **60** is magnetized to facilitate connection to a magnetic charging cord. The flashlight housing **52** seals against the illumination light source **54**, the flashlight switch **58**, and the flashlight charging port **60** to prevent water from entering the flashlight housing **52** and interfere with electrical components.

[0026] Similarly, the laser emitter module **62** comprises a laser emitter housing **64**, an laser light source **66**, a laser emitter power supply **68**, a laser emitter switch **70**, and a laser emitter charging port **72**. The laser emitter power supply **68** comprises a laser emitter battery but may comprise a capacitor or other suitable power source. The laser emitter switch **70** is electrically coupled to the laser emitter power source and the laser light source **66** and is operable to activate the laser light source **66** to emit a visible laser forwardly of the handguard **12** when the laser emitter module **62** is mounted to the handguard **12**. The laser emitter module **62** is usable to aim the firearm **110** when mounted to the handguard **12**. When removed from the handguard **12**, the laser emitter module **62** may be used as a typical laser pointer to indicate remote objects. The laser emitter charging port **72** is mounted to the laser emitter housing **64** and is electrically coupled to the laser emitter power supply **68** to facilitate charging the laser emitter power supply **68**. The laser emitter charging port **72** is magnetized to facilitate connection to a magnetic charging cord. The laser emitter housing **64** seals against the laser light source **66**, the laser emitter switch **70**, and the laser emitter charging port **72** to prevent water from entering the laser emitter housing **64** and interfering with electrical components.

[0027] Alignment members **86** protrude from the flashlight module **50** and the laser emitter module **62** which fit into notches **88** in the flashlight receiver **76** and the laser emitter receiver **82** respectively when the flashlight module **50** and the laser emitter module **62** are mounted to the handguard **12**. The alignment members **86** orient the flashlight module **50** and the laser emitter module **62** such that the flashlight switch **58**, the laser emitter switch **70**, the flashlight charging port **60**, and the laser emitter charging port **72** are not obstructed by the handguard **12** or the monopod **26**.

[0028] With reference to FIG. **6**, an embodiment is shown with a pair of monopods **26** substantially limited as described above. Each monopod **26** is interchangeably couplable to the handguard **12** and includes a tool **94** that is pivotally attached to the monopod **26**. The tool **94** of each monopod **26** is pivotable near the distal end **30** of the monopod **26** between a deployed position **96** and a stored position **98**. In the stored position **98**, the tool **94** is partially stored in the monopod **26** and extends generally from the distal end **30** of the monopod **26** toward the proximal end **28** of the

monopod **26**. When in the deployed position **96**, the tool **94** extends away from the monopod **26** in alignment with a longitudinal axis of the monopod **26**.

[0029] The tool **94** of a first monopod **90** depicted in FIG. **6** is a knife blade **100**. The first monopod **90** covers a cutting edge **102** of the knife blade **100** when it is in the stored position **98**, and the cutting edge **102** faces a same direction as the front surface **32** of the monopod **26** when in the deployed position **96**. The knife blade **100** also has a piercing point **104** which is positioned distally and faces away from the monopod **26** when the knife blade **100** is in the deployed position **96**.

[0030] The tool **94** of a second monopod **92** shown in FIG. **6** is an ice pick **106**. The second monopod **92** covers a pointed end **108** of the ice pick **106** when the ice pick **106** is in the stored position **98**. The pointed end **108** is positioned distally and faces away from the monopod **26** when in the deployed position **96**.

[0031] In use, the monopod **26** may be placed in the extended position **38** to support the firearm **110** on a support surface or orient the firearm **110** via grasping the monopod **26** when the handguard **12** is attached to the firearm **110**. The monopod **26** may be placed in the retracted position **36** to facilitate the user in gripping or cradling the handguard **12** in the user's hand. The flashlight module **50** and the laser emitter module **62** may be used with or separately from the handguard **12** as described above. In embodiments where the monopod **26** has a tool **94** attached thereto, the monopod **26** may be removed and used as a gripping element for the tool **94**. In some cases, it may be advantageous to use the tool **94** while the monopod **26** is attached to the handguard **12**. For example, in embodiments with the knife blade **100**, the knife blade **100** may be positioned in the deployed position **96** and the monopod **26** may be placed in the retracted position **36**. In this manner, the knife blade **100** protrudes forwardly from the handguard **12** to define a bayonet.

[0032] With respect to the above description then, it is to be realized that the optimum dimensional relationships for the parts of an embodiment enabled by the disclosure, to include variations in size, materials, shape, form, function and manner of operation, assembly and use, are deemed readily apparent and obvious to one skilled in the art, and all equivalent relationships to those illustrated in the drawings and described in the specification are intended to be encompassed by an embodiment of the disclosure.

[0033] Therefore, the foregoing is considered as illustrative only of the principles of the disclosure. Further, since numerous modifications and changes will readily occur to those skilled in the art, it is not desired to limit the disclosure to the exact construction and operation shown and described, and accordingly, all suitable modifications and equivalents may be resorted to, falling within the scope of the disclosure. In this patent document, the word “comprising” is used in its non-limiting sense to mean that items following the word are included, but items not specifically mentioned are not excluded. A reference to an element by the indefinite article “a” does not exclude the possibility that more than one of the element is present, unless the context clearly requires that there be only one of the elements.

Claims

1. (canceled)
2. (canceled)
3. (canceled)
4. (canceled)
5. (canceled)
6. (canceled)
7. (canceled)
8. (canceled)
9. (canceled)

10. (canceled)

11. (canceled)

12. (canceled)

13. (canceled)

14. A firearm attachment comprising: a handguard configured to mount to a barrel of a firearm having a magazine extending downwardly relative to the barrel; a monopod coupled to a bottom side of the handguard, the monopod being elongated between proximal and distal ends relative to the handguard such that the distal end of the monopod is configured to extend farther below the barrel of the firearm than the magazine of the firearm, the monopod being movable between an extended position and a retracted position, the monopod extending away from the handguard from the handguard from the proximal end to the distal end when in the extended position; a monopod fastener removably coupling the monopod to the handguard; a retractable tool attached to the monopod, the tool being movable between a stored position and a deployed position, the tool being at least partially housed in the monopod when in the stored position, the tool extending away from the monopod in alignment with a longitudinal axis of the monopod when in the deployed position; and wherein the tool is a knife blade, a cutting edge of the knife blade facing forwardly with respect to the handguard when the monopod is in the extended position and the knife blade is positioned in the deployed position, the knife blade having a piercing end positioned distally and facing away from the monopod when in the deployed position.

15. A firearm attachment comprising: a handguard configured to mount to a barrel of a firearm having a magazine extending downwardly relative to the barrel; a monopod coupled to a bottom side of the handguard, the monopod being elongated between proximal and distal ends relative to the handguard such that the distal end of the monopod is configured to extend farther below the barrel of the firearm than the magazine of the firearm, the monopod being movable between an extended position and a retracted position, the monopod extending away from the handguard from the handguard from the proximal end to the distal end when in the extended position; a monopod fastener removably coupling the monopod to the handguard; a retractable tool attached to the monopod, the tool being movable between a stored position and a deployed position, the tool being at least partially housed in the monopod when in the stored position, the tool extending away from the monopod in alignment with a longitudinal axis of the monopod when in the deployed position; and wherein the tool is an ice pick, the ice pick a pointed end of the ice pick being positioned distally and facing away from the monopod when in the deployed position.

16. A firearm attachment comprising: a handguard having a tubular shape, the handguard having a front end and a back end which are open to an interior space of the handguard, the handguard being shaped and configured to receive a barrel of a firearm having a magazine extending downwardly relative to the barrel into the interior space via the back end and out of the interior space via the front end; a monopod coupled to a bottom side of the handguard, the monopod being elongated between proximal and distal ends relative to the handguard such that the distal end of the monopod is configured to extend farther below the barrel of the firearm than the magazine of the firearm, the monopod being pivotable between an extended position and a retracted position, the monopod extending away from the handguard from the handguard from the proximal end to the distal end when in the extended position, the monopod extending forwardly along the handguard from the proximal end to the distal end when in the retracted position, the distal end tapering to a point, the monopod having a front surface which faces forwardly relative to the handguard when the monopod is in the extended position, the front surface of the monopod defining a plurality of recesses sized and configured to receive fingers of a user therein; a monopod fastener removably coupling the monopod to the handguard, the monopod fastener comprising an insertion member coupled to the handguard and a receiving member coupled to the monopod, the insertion member being receivable into the receiving member, the monopod fastener further comprising a latch member coupled to the monopod, the latch member being biased to interlock with the insertion

member; a flashlight module mounted to the handguard, the flashlight module comprising: a flashlight housing; an illumination light source coupled to the flashlight housing, the illumination light source being oriented to emit light forwardly of the handguard; a flashlight power supply mounted in the flashlight housing, the flashlight power supply being electrically couplable to the illumination light source, the flashlight power supply comprising a flashlight battery; a flashlight switch mounted to the flashlight housing and electrically coupled to the flashlight power supply and the illumination light source, the flashlight switch being operable to electrically couple the flashlight power supply to the illumination light source to activate the illumination light source; and a flashlight charging port mounted to the flashlight housing and electrically coupled to the flashlight power supply; a flashlight mount coupled to the handguard, the flashlight module being removably coupled to the handguard via the flashlight mount, the flashlight mount comprising a flashlight receiver which defines a flashlight-receiving cavity for receiving the flashlight module therein, the flashlight mount further comprising a flashlight magnet mounted to the flashlight module to secure the flashlight module in the flashlight receiver; a laser emitter module mounted to the handguard, the laser emitter module comprising: a laser emitter housing; a laser light source coupled to the laser emitter housing, the laser light source being oriented to emit a visible laser forwardly of the handguard; a laser emitter power supply mounted in the laser emitter housing, the laser emitter power supply being electrically couplable to the illumination light source, the laser emitter power supply comprising a laser emitter battery; a laser emitter switch mounted to the laser emitter housing and electrically coupled to the laser emitter power supply and the laser light source, the laser emitter switch being operable to electrically couple the laser emitter power supply to the laser light source to activate the laser light source; and a laser emitter charging port mounted to the laser emitter housing and electrically coupled to the laser emitter power supply; and a laser emitter mount coupled to the handguard, the laser emitter module being removably coupled to the handguard via the laser emitter mount, the laser emitter mount comprising a laser emitter receiver which defines a laser emitter-receiving cavity for receiving the laser emitter module therein, the laser emitter mount further comprising a laser emitter magnet mounted to the laser emitter module to secure the laser emitter module in the laser emitter receiver; wherein the flashlight mount and the laser emitter mount are positioned opposite each other across the monopod; and one of a knife blade and an ice pick being attached to the monopod, wherein the knife blade is movable between a stored position and a deployed position, the knife blade being at least partially housed in the monopod when in the stored position, a cutting edge of the knife blade being covered by the monopod when in the stored position, the knife blade extending away from the monopod in alignment with a longitudinal axis of the monopod when in the deployed position, the cutting edge of the knife blade facing a same direction as the front surface of the monopod when in the deployed position, the knife blade having a piercing end positioned distally and facing away from the monopod when in the deployed position, and the ice pick is movable between a stored position and a deployed position, the ice pick being at least partially housed in the monopod when in the stored position, a pointed end of the ice pick being covered by the monopod when in the stored position, the ice pick extending away from the monopod in alignment with a longitudinal axis of the monopod when in the deployed position, the pointed end of the ice pick being positioned distally and facing away from the monopod when in the deployed position.

17. (canceled)

18. (canceled)
